

Public-Data Reproduction of GIGA-Lens Modeling for the DESI Strong Lens Foundry Demonstration System DESI-165.4754-06.0423: Final Report*

Greg Benson[†]

Xiaosheng Huang[‡]

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Final results and lessons; the complete development record is evolution.pdf

Abstract

We report the final state of an end-to-end, public-data reproduction of the GIGA-LENS modeling demonstration for the DESI Strong Lens Foundry I system DESI-165.4754-06.0423 (Huang et al., 2025). On the corrected data product — *HST*/WFC3-IR F140W at native scale, with interloper masks, a model-subtracted noise calibration, and a convention-correct PSF kernel — preconditioned HMC at the published compute scale gives $\gamma_{\text{EPL}} = 1.433 [1.400, 1.468]$, consistent with the published 1.372 ± 0.023 at 1.5σ of the combined statistical uncertainties, with $\theta_{\text{E}} = 2.6551$ within 0.33% of the published value, strict convergence ($\hat{R}_{\text{max}} = 1.077$, γ_{EPL} ESS = 5,714), and the published inner-critical-curve topology recovered. Beyond the statistical error (± 0.034), the slope carries two systematics of comparable size: a ~ 0.1 model-class term (source/companion flexibility) and a ~ 0.1 data-scale term (a low/steep-slope bimodality that exists only on upsampled drizzle products and would require a correlated-noise likelihood to resolve). Reaching this end state surfaced two findings of direct relevance to the GIGA-LENS stack: simulator PSF kernels must be sampled at the data pixel scale (a convention violation broadened our effective PSF $2\times$ and was the entire native-scale misfit floor), and sky-based noise calibrations are biased by diffuse lens-wing flux unless calibrated on model-subtracted residuals. We provide the final parameter comparison, the corrected data/inference recipe, the uncertainty budget, seven documented upstream issues with fixes, and the compute cost (44 of 200 budgeted A100-hours).

1 Introduction

Huang et al. (2025) demonstrated GPU-accelerated Bayesian lens modeling with GIGA-LENS (Gu et al., 2022) on the Foundry I system DESI-165.4754-06.0423, reporting a full 74-parameter posterior (EPL + external shear mass model, four-Sérsic lens light, Sérsic + shapelet source) from MAP $\rightarrow$ SVI $\rightarrow$ preconditioned HMC in 193 minutes on one Perlmutter A100 node. This document is the **final report** of an independent, public-data reproduction of that analysis: it presents the

*All work reported here — the code, experiments, analysis, and text — was performed by Claude Code (Anthropic), running the Claude Fable 5 model, guided by Greg Benson.

[†]benson@cs.usfca.edu; Department of Computer Science, University of San Francisco, CA 94117, USA

[‡]xhuang@usfca.edu; Department of Physics & Astronomy, University of San Francisco, CA 94117; and Physics Division, Lawrence Berkeley National Laboratory, CA 94720

end state — the corrected data treatment, the final posterior and its uncertainty budget, and what the exercise established about the GIGA-LENS code base — and omits the development history.

The reproduction was developed over thirteen internal versions (May–June 2026), including an extended investigation of HMC pathologies, a paper-scale campaign on NERSC Perlmutter, and a group-review-driven rigor revision that retired several intermediate claims. That complete record, including every superseded result and the reasoning that led to each correction, is preserved verbatim in the companion document `evolution.pdf` (<https://github.com/usfcs-edu/agentic-lensing/blob/main/reproductions/foundry-i/papers/evolution.pdf>); the campaign execution log is `PERLMUTTER.CAMPAIGN.md` in the repository. Here, superseded claims appear only as named retractions (§7).

2 Headline Result and Uncertainty Budget

2.1 The final posterior

Table 1 compares the final posterior — preconditioned HMC with $48 \text{ chains} \times 8,000 \text{ draws}$ on `cutout.v2d`, the corrected native-scale data product defined in §3 — with Huang et al. (2025, Table 3). Every mass parameter is strictly converged ($\hat{R}_{\max}$ over all 74 parameters = 1.077; minimum ESS over all parameters = 1,355). The posterior-median model reproduces the tangential *plus inner* critical-curve topology that the published Fig. 8 identifies as the signature of $\gamma_{\text{EPL}} < 2$ (Figure 1), with a masked reduced residual of $\chi^2_\nu = 1.26$ against a sky-calibrated value of 1.00.

Table 1: Final mass-model posterior versus the published values. This work: median and [16, 84]% interval from $48 \times 8,000$ preconditioned-HMC draws on `cutout.v2d`.

Parameter	This work	$\hat{R}$	ESS	Huang+2025a	Note
θ_{E} (″)	2.6551 [2.6530, 2.6572]	1.001	63,578	2.6463 ± 0.0017	+0.33%
γ_{EPL}	1.4330 [1.3995, 1.4685]	1.016	5,714	1.372 ± 0.023	1.5σ comb.; §2.2
e_1	0.1070 [0.1045, 0.1096]	1.014	7,024	0.1091 ± 0.0020	−1.9%
e_2	+0.0835 [0.0807, 0.0864]	1.005	24,114	-0.1320 ± 0.0020	frame; §2.3
$\gamma_{\text{ext},1}$	+0.0710 [0.0684, 0.0736]	1.004	27,338	$+0.0657 \pm 0.0024$	+8.1%
$\gamma_{\text{ext},2}$	+0.0399 [0.0365, 0.0432]	1.010	9,393	-0.0939 ± 0.0022	frame; §2.3

Note. $\hat{R}_{\max}$ over all 74 parameters = 1.077; $\text{ESS}_{\min}$ over all parameters = 1,355. The e_2 and $\gamma_{\text{ext},2}$ rows differ in sign from the published catalog because the two reductions use mirrored position-angle frames; compared in polar form the orientations agree to 6° (§2.3).

2.2 Uncertainty budget for γ_{EPL}

The statistical error on the slope is ± 0.034 . Two systematics of comparable or larger size must be quoted with it:

- **Model class** (~ 0.1). Raising the source shapelet order to $n_{\max} = 8$ and adding a third companion Sérsic improves the MAP only marginally ($\chi^2_\nu = 1.234 \rightarrow 1.203$, not plateaued) but moves the slope from 1.41 to 1.31. The published value (1.372) lies inside this band.
- **Data scale** (~ 0.1). The low-slope solution moves from $\gamma_{\text{EPL}} = 1.29$ (binned 0″08 product) through 1.43 (native scale) with the published 1.37 (at 0″065) between them. A sharper

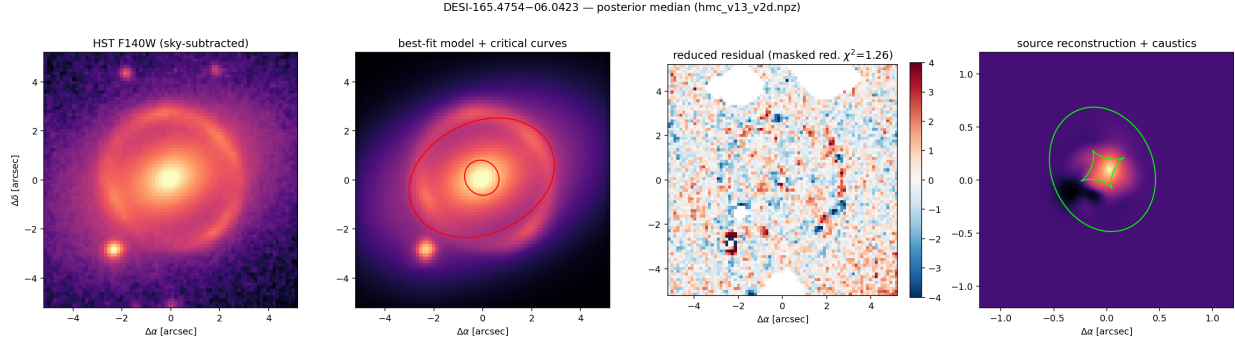


Figure 1: Fig.-8 analogue at the final posterior median ($\gamma_{\text{EPL}} = 1.433$): observed image, best-fit model with critical curves, masked reduced residual ($\chi^2_\nu = 1.26$), and source reconstruction with caustics. The inner critical curve and caustic — the feature the published Fig. 8 calls out for $\gamma_{\text{EPL}} < 2$ — are recovered.

version of the same scale dependence appears as an explicit bimodality on upsampled products (§8).

The defensible summary statement is therefore $\gamma_{\text{EPL}} = 1.43 \pm 0.03$ (stat) ± 0.1 (syst), in agreement with the published 1.372 ± 0.023 , whose own error bar is purely statistical.

2.3 The orientation frame of e_2 and $\gamma_{\text{ext},2}$

The signed e_2 and $\gamma_{\text{ext},2}$ disagree with the published catalog while e_1 and $\gamma_{\text{ext},1}$ agree — the signature of mirrored position-angle conventions rather than a physics discrepancy. In polar form our posterior gives $|e| = 0.136$ at mass position angle $+19.0^\circ$ and $|\gamma_{\text{ext}}| = 0.081$ at $+14.7^\circ$ (in our pixel frame), versus the published $|e| = 0.171$ at -25.2° and $|\gamma_{\text{ext}}| = 0.115$ at -27.5° (in theirs). Three facts pin this down as a frame difference. First, the deflector *light*, measured directly from second moments of the image we model, lies at $+25.5^\circ$: our mass is aligned with the light to 6° , as expected physically, whereas the published angle interpreted in our frame would be 50° from the light. Second, under a parity flip (PA $\rightarrow$ $-$ PA) the published mass angle maps to $+25.2^\circ$ — within 0.3° of our measured light angle. Third, the two reductions model different image products (we model the MAST HAP mosaic; the paper drizzled its own stack), so a mirrored axis convention between them is unremarkable. After the reflection is removed, the genuine residuals are a 6.2° rotation of the mass ellipticity, a 12.8° rotation of the shear, and 21%/29% smaller $|e|/|\gamma_{\text{ext}}|$ amplitudes — a coherent displacement along the well-known ellipticity–shear coupling, of a size consistent with the ± 0.1 -class systematics above.

3 Data and Corrected Treatment

DESI-165.4754-06.0423 was observed with *HST*/WFC3-IR in F140W under program GO-15867 (PI Huang): 1197.7 s total in three exposures, public via MAST DOI 10.17909/hx0v-9260. We model the high-level pipeline-combined drizzle product `hst_15867_65_wfc3_ir_f140w_ie5065_drz.fits` at its true scale of 0.12825 pix^{-1} (the D001SCAL keyword; the WFC3-IR nominal 0.13 is a 1.36% relabel), and, for cross-scale checks, the MAST HAP fine-skycell mosaic of the same exposures at 0.04 pix^{-1} and a 2×2 -binned 0.08 rebuild of it.

The final data product (`cutout_v2d`, built by `40_make_cutout_v2.py` + `40d_fix_v2_psf.py`) applies three corrections that proved decisive:

- **Crop and masks.** An 80×80 px ($10''4$) crop so the lens fills the frame; whole-object masks on the three faint field galaxies; a mask on the compact interloper superposed on arc A (identified automatically as the most point-like member of the closest ring-detection pair, leaving all four lensed images unmasked); and $r \leq 0''.2$ core masks on the lens and companion centers (Figure 2).
- **Noise calibrated on model-subtracted sky.** The per-pixel error map is the drizzle WHT-implied sigma plus a data-based Poisson term. The sky-term normalization is calibrated so that *model-subtracted* residuals in the source-free annulus have unit reduced χ^2 : calibrating on raw image fluctuations — the common shortcut — is biased high by diffuse lens-wing flux that the model fits (21% in variance at native scale, $3.3\times$ on the fine skycell). With the wing bias removed, the WHT-implied noise is correct at native scale to 1%. The Poisson term is verified rather than assumed: removing it overshoots the arc-region χ^2 to 2.26, so it is kept.
- **Convention-correct PSF (§4).**

Product tags used throughout: **v2c** = native scale with the PSF fix only (noise as before; isolates the PSF effect); **v2d** = v2c + the model-subtracted noise calibration (the final product); **v3** = the $0''.04$ fine-skycell cutout; **v3b** = its 2×2 -binned $0''.08$ rebuild.

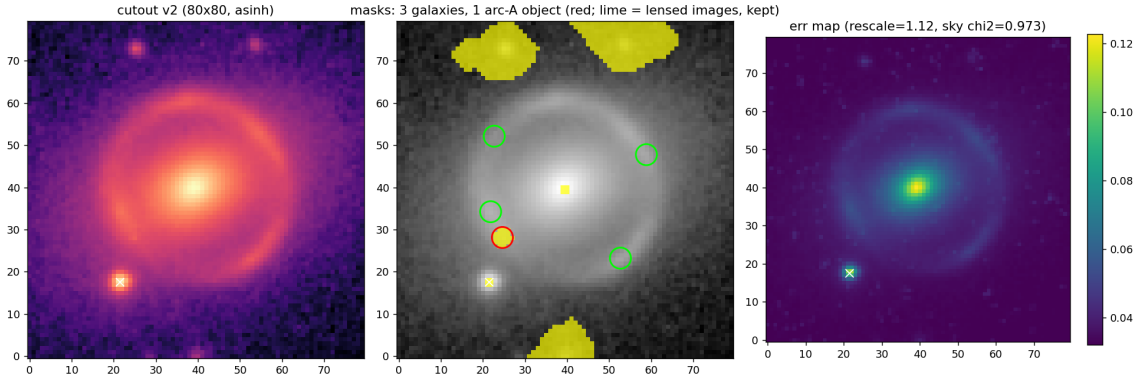


Figure 2: The final native-scale data product: sky-subtracted cutout, keep-mask (three field galaxies, the arc-A interloper, and the two galaxy cores masked), and the sky-calibration region.

4 The PSF: Construction and the Sampling-Convention Rule

4.1 Empirical PSF

The GIGA-LENS-default TinyTim F140W kernel is too compact for this analysis: it is identically zero beyond $\pm 0''.42$ where the real PSF carries measurable power, and replacing it with an empirical PSF moves the global-best MAP basin into the published parameter quadrant. Our kernel is built with `photutils.psf.EPSFBuilder` (Bradley et al., 2024) from 30 isolated in-field stars of the same drizzle ($2\times$ oversampling, quartic smoothing, eight outer iterations), cropped to 27×27 supersampled pixels ($\pm 0''.85$) and unit-normalized (`17_build_empirical_psf.py`; a native-sampled 51×51 variant from 25 skycell stars serves the fine-scale products). An in-exposure empirical PSF should be considered a default ingredient for this kind of modeling, not an enhancement.

4.2 The sampling-convention rule

The single largest defect found in this reproduction was not astrophysical: GIGA-LENS’s simulator passes `sim_config.kernel` through LENSSTRONOMY’s `subgrid_kernel`, which **assumes the kernel is sampled at the data pixel scale** (`delta_pix`) and upsamples it internally by the supersampling factor. Supplying our $0''.065$ -sampled empirical kernel together with `delta_pix = 0.13` and `supersample = 2` therefore stretched the effective PSF by $2\times$ (r_{50} ratio 1.92; model images change by 5.4σ RMS when corrected). Every native-scale fit of the development record ran with this broadened kernel.

Refitting the same data with a convention-correct kernel (a $0''.13$ -sampled 15×15 kernel least-squares-fitted through `subgrid_kernel` itself) and *no other change* drops the native-scale MAP floor from $\chi^2_{\nu} = 3.42\text{--}3.55$ to 1.051 (Figure 3; Table 2). It also resolves what had appeared to be a sampler limitation: the slope’s effective sample size, stuck at 259 after 13,000 draws per chain under the broadened kernel, reaches 5,714 from 8,000 draws once the PSF is corrected — the broadened kernel, not the sampler, had flattened the γ_{EPL} direction.

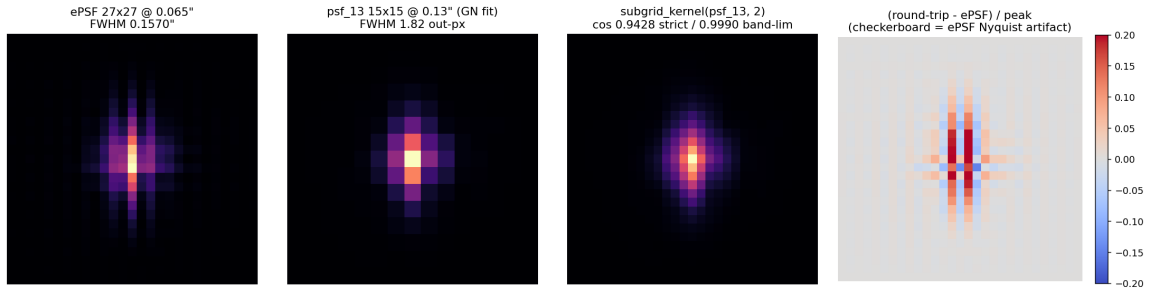


Figure 3: The PSF sampling-convention fix: the $0''.065$ -sampled empirical kernel (left) versus the $0''.13$ -sampled kernel fitted through the simulator’s own `subgrid_kernel` operator, and the effective PSF each produces in the forward model. The defective configuration broadens the effective PSF by $2\times$ in r_{50} .

5 Lens Model

We adopt the published parametric model (Huang et al., 2025): a single elliptical power-law convergence (EPL; Birrer and Amara 2018; Gu et al. 2022) with Einstein radius θ_E , slope γ_{EPL} , ellipticity (e_1, e_2) and centroid, plus external shear ($\gamma_{\text{ext},1}, \gamma_{\text{ext},2}$) — 8 mass parameters; four elliptical Sérsic profiles for the lens light (two for the deflector, two for the companion galaxy at $(-2''.34, -2''.86)$); and a Sérsic + Cartesian-shapelet source ($n_{\text{max}} = 6$, 28 amplitude coefficients with $\mathcal{N}(0, 5/\sqrt{i+1})$ priors). Total: 74 free parameters; the full prior table is Appendix A.

Two implementation choices matter. First, `use_lstsq=False` everywhere: GIGA-LENS’s unconstrained least-squares pass for linear amplitudes can drive Sérsic intensities negative and absorb χ^2 at non-physical configurations (it degraded Einstein-radius accuracy from 0.5% to 43% in early ablations), and its `pinv`-based profiled posterior is non-smooth (§9). Second, the central $r \leq 0''.2$ degeneracy mask is implemented by inflating the error map inside the mask — the smallest change to GIGA-LENS that yields a masked likelihood, since `ForwardProbModel` exposes no mask argument.

6 Inference: the Recipe That Works at Published Quality

The published pipeline recipe — multi-start MAP, full-covariance Gaussian SVI, preconditioned HMC with momentum covariance Σ_{VI}^{-1} — works as advertised once given adequate scale and the corrected inputs above. The configuration that reaches published quality (one Perlmutter node, 4×A100, vendored `gigalens-sean multinode-2025` library):

- **MAP:** 1,000 particles × 6,000–8,000 AdaBelief steps with a polynomial learning-rate decay and a loss-plateau stop; cold and warm starts agree. Runtime 8–26 min.
- **SVI:** 1,000 particles, 15,000 steps, `init_scales` = 3×10^{-3} , quadratic learning-rate ramp. The longer schedule and larger `init_scales` are what produce a *full-rank* 74×74 variational covariance (a 1,500-step run yields rank 53/74, which silently breaks the HMC preconditioner; §9). ELBO must be smooth and flattened. Runtime 14–55 min depending on product size.
- **HMC:** 48 chains from SVI draws, ChEES trajectory adaptation, dual-averaging step size, momentum covariance = Σ_{VI}^{-1} ; 750 burn-in + 8,000 draws in 10 min.
- **Environment:** single or few devices with `XLA_FLAGS=--xla_gpu_autotune_level=0`; never run this likelihood on CPU (a grouped-convolution XLA pathology makes it $216\times$ slower than a single L4).

For HMC work outside the `ModellingSequence` path we also marginalize the 28 shapelet amplitudes analytically, using their own Gaussian priors as a Tikhonov ridge: with W the per-pixel precision, X the lensed, PSF-convolved unit-amplitude design matrix, and R the residual after the deterministic components,

$$A = X^\top W X + \Lambda, \quad \Lambda_{ii} = \frac{i+1}{25},$$

$$a^\star = \text{cho_solve}(A, X^\top (W \odot R)), \quad \log L = -\frac{1}{2} \sum W R^2 + \frac{1}{2} b^\top a^\star - \frac{1}{2} \log \det(A),$$

where the ridge makes A strictly positive-definite (Cholesky, never `pinv`) and $-\frac{1}{2} \log \det A$ is the Gaussian-evidence (Occam) term that GIGA-LENS’s `lstsq_simulate` omits. This single change reduced the gradient floor of the profiled posterior by $228\times$ and is the fix we recommend upstream (§9).

Tables 2 and 3 summarize the end state of the fit quality and sampling diagnostics across data products.

7 Why Earlier Attempts Disagreed: the Defect Chain

The development record’s central discrepancy — a 57%-steeper slope ($\gamma_{\text{EPL}} = 2.15$) that resisted 119 GPU-hours of sampling effort — dissolved entirely under five corrections, none of them model physics:

1. **Uncropped field:** modeling a 128^2 cutout left the fit free to spend flexibility on field structure; the corrected product crops to 80^2 so the lens fills the frame.
2. **Unmasked interlopers:** three faint galaxies and a compact object superposed on arc A were previously fit rather than masked.

Table 2: Final χ^2 state across data products (masked reduced χ^2 at the MAP, 1,000-particle paper-scale fits, all plateaued unless noted).

configuration	data product	χ^2_ν
legacy floor (broadened PSF)	native 80 ²	3.42–3.55
PSF convention fixed, same noise	native (v2c)	1.051
+ model-subtracted noise calibration	native (v2d)	1.234
flexibility ablation ($n_{\max}=8$, comp. 3; not plateaued)	v2d	1.203
fine skycell, recalibrated σ	260 ² @ 0''04 (v3)	0.915
binned rebuild, low-slope basin	130 ² @ 0''08 (v3b)	1.594
binned rebuild, steep basin	v3b	1.516

Note. The residual native-scale floor above 1.0 localizes at the companion galaxy ($\chi^2 \simeq 9$ over its 0''8 region), which the flexibility ablation only marginally improves — a real structure limitation of the two-Sérsic companion treatment, not of the lens model.

Table 3: Sampling diagnostics at the final state. The published bar is $\hat{R} < 1.10$, ESS 32,200–40,000.

run	draws/chain	γ_{EPL} median [16, 84]%	θ_{E}	$\hat{R}_{\max}$	ESS _{min}
v2d (final)	8,000	1.433 [1.400, 1.468]	2.6551	1.077	5,714
v3 (0''04)	3,500	2.585 [2.564, 2.605]	2.5927	1.007	28,124
v3b low basin	3,500	1.293 [1.283, 1.305]	2.6033	—	—
published	750	1.372 $\pm$ 0.023	2.6463	< 1.10	32,200

Note. The v3 row samples at published efficiency but its steep-slope posterior reflects the upsampled product’s mis-weighted independent-pixel likelihood (§8). The v3b row is conditional on the low-slope basin (45 of 48 chains; the other 3 sample the steep mode with zero inter-basin migrations, so cross-chain $\hat{R}$ /ESS are deliberately not quoted).

- Under-invested optimization:** early MAP/SVI used $\sim 25\times$ fewer particle-steps than the published runs; at paper scale the same code converges cleanly.
- Wing-contaminated noise calibration:** sky-based σ calibrations on raw image fluctuations were inflated by diffuse lens-wing flux. Two intermediate claims are retracted accordingly: an apparent “WHT underestimates sky noise by 25%” finding (rescale 1.117) was an artifact of calibrating on wing-contaminated raw sky, not a WHT defect — recalibrated on model-subtracted residuals (variance factor 0.787), the drizzle-weight noise is correct to 1% ($1.117 \times \sqrt{0.787} = 0.99$); and a fine-scale “ $\chi^2_\nu = 0.451$ ” quoted as meeting the fit-quality bar (recalibrated: 0.915 — which *does* meet it, with a calibrated statistic; Figure 4).
- The PSF sampling-convention defect** (§4.2): the entire remaining native-scale misfit floor, and the apparent γ_{EPL} -mixing pathology, were the $2\times$ -broadened kernel.

An earlier hypothesis that the residual parameter gaps indicated a missing physical constraint (a mass–light position-angle tie or a higher-order multipole) is retracted: the published model contains neither — its Table 2 priors are fully independent — and the gap closed through items 1–5 alone. Likewise, soft posterior directions once labeled “mass-sheet / slope–ellipticity degeneracies” are more honestly described as a broad valley coupling γ_{EPL} , e_1/e_2 , and γ_{ext} under the pre-correction likelihood; the valley largely closed once the likelihood was fixed.

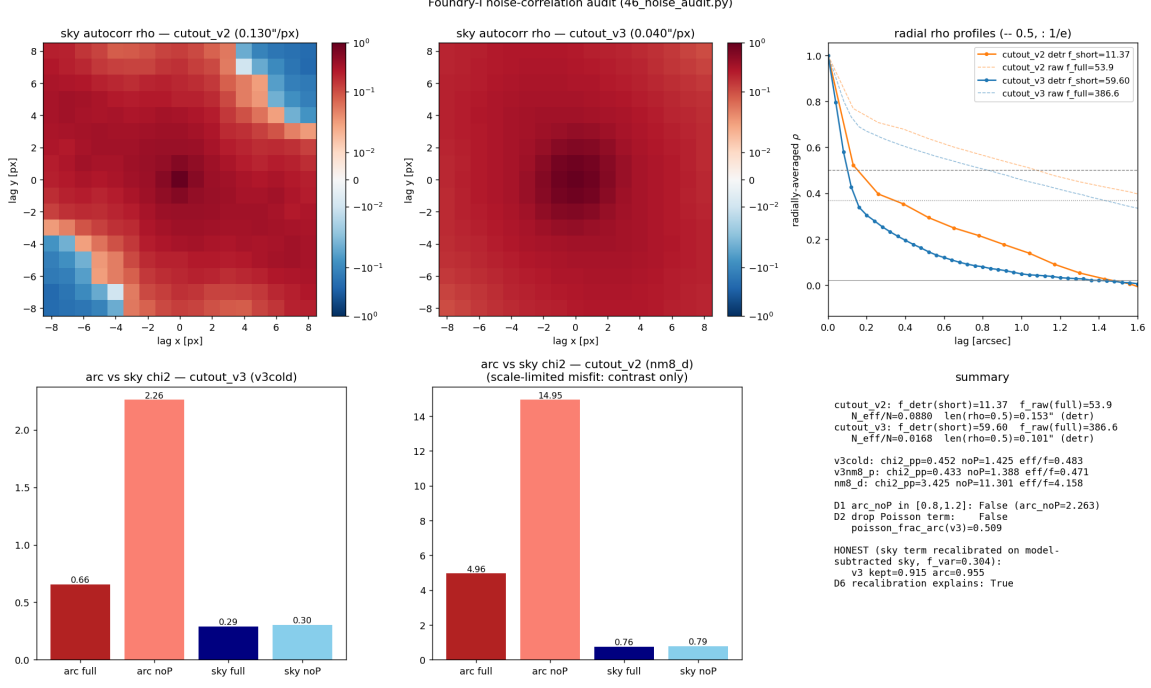


Figure 4: The noise-calibration audit behind defect 4: sky autocorrelation (raw versus background-detrended), arc- versus sky-region χ^2 with and without the Poisson term, and the model-subtracted (“honest”) recalibration that resolves the fine-scale $\chi^2_\nu = 0.451$ artifact to 0.915.

8 Open Findings

Scale-dependent slope bimodality. On drizzle products upsampled beyond the native scale, a steep-slope solution ($\gamma_{\text{EPL}} \approx 2.4\text{--}2.7$) coexists with the low-slope one. On the 2×2 -binned $0''.08$ product the steep basin retains a $\Delta\chi^2_\nu = 0.078$ likelihood preference even after the corrected calibration, and HMC demonstrates the disconnection directly: 45 of 48 chains sample the low mode, 3 the steep mode, with zero inter-basin migrations in 3,500 draws. At native scale the steep mode is not even metastable — a MAP started from the fine-scale steep solution drains monotonically to $\gamma_{\text{EPL}} \approx 1.17$. The bimodality is therefore a property of upsampled products (Figure 5 shows the binned-product low-basin fit), most plausibly through the interaction of the mis-weighted independent-pixel likelihood with sub-native-pixel structure.

Pixel correlation, quantified. Background-detrended sky autocorrelation of the $0''.04$ skycell gives a correlation length of 2.5 fine pixels — the $3.2\times$ upsampling footprint — and direct binning shows 2×2 fine-pixel blocks are $\sim 78\%$ internally correlated. The fine grid carries between $\sim 1/10$ and $\sim 1/60$ as many independent elements as pixels (convention-dependent), so per-pixel statistics on upsampled products overstate the information content even when the marginal variance is calibrated.

The methodological need. Both findings point at the same missing ingredient: a correlated-noise (pixel-covariance) likelihood. With it, the fine- and native-scale posteriors should unify and the bimodality should resolve; without it, native-scale modeling is the defensible choice, which is why the headline posterior comes from `cutout_v2d`.

Companion structure. The residual native-scale floor ($\chi^2_\nu = 1.23$ versus the sky-calibrated 1.0) localizes at the companion galaxy and is only marginally improved by added Sérsic flexibility; a structurally richer companion model is the natural next increment.

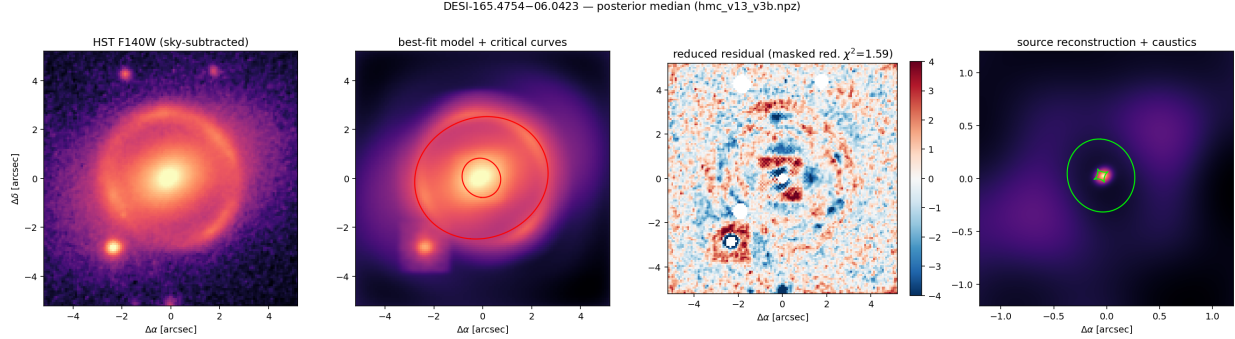


Figure 5: The binned ($0''.08$) product at its low-basin posterior median ($\gamma_{\text{EPL}} = 1.29$): the inner critical curve persists, but the binned likelihood retains a $\Delta\chi^2_\nu = 0.078$ preference for a steep-slope mode that does not survive at native scale — the sharpest expression of the open correlated-noise question.

9 Guidance for the GIGA-Lens Code Base

Seven issues in the GIGA-LENS 2.0 / JAX 0.6.2 / TFP 0.25.0 stack were documented at a reproducible level during this work (full detail in `evolution.pdf`, §*Upstream issues*):

1. **Implicit `shard_map` import** — attribute access without importing `jax.experimental.shard_map` fails under JAX 0.6.2; callers must pre-import it.
2. **Negative lens-light amplitudes via unconstrained `lstsq`** — the least-squares amplitude path can return unphysical negative intensities that absorb χ^2 ; `use_lstsq=False` (or an NNLS variant upstream) avoids it.
3. **The multi-hour HMC “compile”** is a `pmap`×grouped-convolution×cuDNN-autotuner pathology, not compilation: single/few devices with `--xla_gpu_autotune_level=0` compile the same kernel in 31–38 s.
4. **`lstsq_simulate` omits the Gaussian-evidence term and is unregularized**, making the profiled posterior non-smooth on near-rank-deficient shapelet design matrices. The fix (§6) is a prior-derived Tikhonov ridge plus the $-\frac{1}{2} \log \det$ term.
5. **Mixed-GPU-type `shard_map` crash** on heterogeneous hosts; UUID-pinning is the workaround.
6. **`jnp.linalg.cholesky` silently returns NaN** on a rank-deficient SVI covariance, producing stuck-looking chains; defensive jitter (or a NaN-trace warning) is needed wherever an SVI covariance feeds a TFP preconditioner.
7. **`DualAveragingStepSizeAdaptation` collapses** on multimodal-Hessian targets; windowed mass-matrix adaptation would require a `JointDistribution` refactor of the prob model.

Beyond bug-level issues, three integration rules emerged that belong in GIGA-LENS documentation (or asserts): (a) the PSF kernel passed to the simulator must be sampled at `delta_pix`

— `subgrid_kernel` upsamples it internally, and a supersampled-scale kernel silently broadens the model PSF (§4.2); (b) SVI must be run long enough, with large enough `init_scales`, to produce a full-rank covariance before HMC preconditioning — rank deficiency fails silently through issue 6; (c) noise maps calibrated from sky fluctuations should use model-subtracted residuals, or diffuse lens light biases the likelihood weight (§7, item 4). An in-exposure empirical PSF and a multi-start MAP with basin reporting are reasonable default additions to the published recipe.

10 Reproduction Recipe, Cost, and Software

The final-state pipeline is six scripts plus Slurm templates in `reproductions/foundry-i/`: `40_make_cutout_v2.py` / `40b-40d` (data products), `41_map_paper_scale.py`, `42_svi_paper_scale.py`, `43_hmc_paper_scale.py`, `44_diagnostics.py`, `45_fig8_panel.py`, and `46_noise_audit.py`, running against a vendored snapshot of the group’s production library (`seanxuseanxu/gigalens`, branch `multinode-2025`, ref `58ec9a7`, in `vendor/`). Paper-scale runs use one Perlmutter GPU node (4×A100); the complete campaign — including the development detours recorded in `evolution.pdf` — consumed **44.1 of 200 budgeted A100-hours**, with the final-state pipeline itself (MAP + SVI + HMC on `cutout_v2d`) costing ~ 3 A100-hours. The *HST* data are public (MAST DOI 10.17909/hx0v-9260); every figure and table in this report is regenerable from the scripts and the job log (`PERLMUTTER.CAMPAIGN.md`).

Table 4: Software stack at the final state.

Package	Version
Python	3.13
JAX	0.6.2 (jax[cuda12])
TensorFlow Probability	0.25.0
GIGA-LENS (modeling library)	<code>seanxuseanxu/gigalens</code> @ <code>multinode-2025</code> (<code>58ec9a7</code>), vendored
LENSTRONOMY	1.14.0
PHOTUTILS	3.0.0
OPTAX	0.2.8
NUMPY / ASTROPY	2.4.6 / current LTS
Production hardware	NERSC Perlmutter, 1 node, 4×A100 40 GB
Development hardware	8×A16 + 2×L4 (local)

11 Conclusions

1. The published Foundry-I analysis reproduces from public data at published sampling quality: θ_E to 0.33%, γ_{EPL} consistent at 1.5σ of the combined statistical uncertainties ($1.433 [1.400, 1.468]$ vs. 1.372 ± 0.023), mass orientation aligned with the measured deflector light to 6° (modulo a mirrored PA frame between the two reductions), and the inner-critical-curve topology recovered.
2. The honest error on the slope is systematics-dominated: ± 0.03 statistical, $\sim \pm 0.1$ model-class, $\sim \pm 0.1$ data-scale. Statistical-only error bars understate the uncertainty of this kind of single-band analysis by a factor of a few.
3. Every discrepancy encountered along the way — including a 57%-steeper slope that once looked like missing model physics — traced to data treatment or software convention: crop, masks,

optimization scale, wing-contaminated noise calibration, and a PSF sampling-convention defect. The corrections are all recipe-level and are documented here and in the scripts.

4. Two contributions are directly actionable upstream: the PSF kernel-sampling rule (with the 2 $\times$ -broadening failure mode as motivation) and the regularized exact marginalization for linear amplitudes (ridge + evidence term), alongside seven documented stack issues.
5. One methodological item remains open: a correlated-noise likelihood for drizzled products, which would unify the scale-dependent posteriors and resolve the upsampled-product slope bimodality.

Acknowledgments

We thank the DESI Strong Lens Foundry team for the feedback that drove the corrected data treatment and the rigor revision. Computations used NERSC Perlmutter (allocation `deepsrch_g`) and local USF hardware. All work reported here — the code, experiments, analysis, and text — was performed by Claude Code (Anthropic), running the Claude Fable 5 model, guided by Greg Benson. *Software:* GIGA-LENS (Gu et al., 2022), LENSTRONOMY (Birrer and Amara, 2018), PHOTUTILS (Bradley et al., 2024), JAX, TensorFlow Probability, OPTAX, NUMPY, ASTROPY.

A Prior Specification

Table 5 gives the prior distributions, closely following Huang et al. (2025, Table 2) with the modifications discussed in §5.

B Side-by-Side with the Published Analysis

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Table 5: Prior specification on the 74 free non-linear parameters.

Parameter group / parameter	Distribution	Parameters
<i>Mass (EPL + external shear; 8 params)</i>		
θ_E	LogNormal	$\mu = \log 2.5, \sigma = 0.25$
γ_{EPL}	TruncatedNormal	$\mu = 2.0, \sigma = 0.25, [1.0, 2.7]$
e_1^m, e_2^m	Normal	$\mu = 0, \sigma = 0.1$
$c_x^{\text{mass}}, c_y^{\text{mass}}$	Normal	$\mu = 0, \sigma = 0.02 \text{ arcsec}$
$\gamma_{\text{ext},1}, \gamma_{\text{ext},2}$	Normal	$\mu = 0, \sigma = 0.05$
<i>Lens light — main lens ($2 \times \text{Sérsic}$, 14 params)</i>		
R_{eff} (compact / extended)	LogNormal	$\mu = \log 0.4 / \log 2.0, \sigma = 0.3$
n_S	Uniform	$[0.5, 8.0]$
$e_{1,2}^{\text{light}}$	TruncatedNormal	$\mu = 0, \sigma = 0.15, [-0.4, 0.4]$
c_x, c_y	Normal	$\mu = 0, \sigma = 0.02 \text{ arcsec}$
I_e (compact / extended)	LogNormal	$\mu = \log 5.0 / \log 2.0, \sigma = 0.5$
<i>Lens light — nearby companion ($2 \times \text{Sérsic}$, 14 params)</i>		
R_{eff}	LogNormal	$\mu = \log 0.3, 0.6, \sigma = 0.3$
c_x, c_y	Normal	$\mu = (-2.34, -2.86), \sigma = 0.1 \text{ arcsec}$
I_e	LogNormal	$\mu = \log 1.0, 0.5, \sigma = 0.5$
e, n	As main lens	—
<i>Source Sérsic (7 params)</i>		
R_{eff}	LogNormal	$\mu = \log 0.5, \sigma = 0.3$
n_S	Uniform	$[0.5, 6.0]$
$e_{1,2}^{\text{src}}$	TruncatedNormal	$\mu = 0, \sigma = 0.15, [-0.5, 0.5]$
c_x, c_y	Normal	$\mu = 0, \sigma = 0.1 \text{ arcsec}$
I_e	LogNormal	$\mu = \log 2.0, \sigma = 0.5$
<i>Source shapelets ($n_{\text{max}} = 6, 1 + 2 + 28 \text{ params}$)</i>		
β	LogNormal	$\mu = \log 0.1, \sigma = 0.1$
c_x, c_y	Normal	$\mu = 0, \sigma = 0.05 \text{ arcsec}$
$\text{amp}_i, i = 0, \dots, 27$	Normal	$\mu = 0, \sigma = 5/\sqrt{i+1}$

Note. The shapelet amplitude scale follows the GIGA-LENS shapelets demonstration; the companion-light centroid prior centers on its detection in early residual maps. Total: 74 free non-linear parameters.

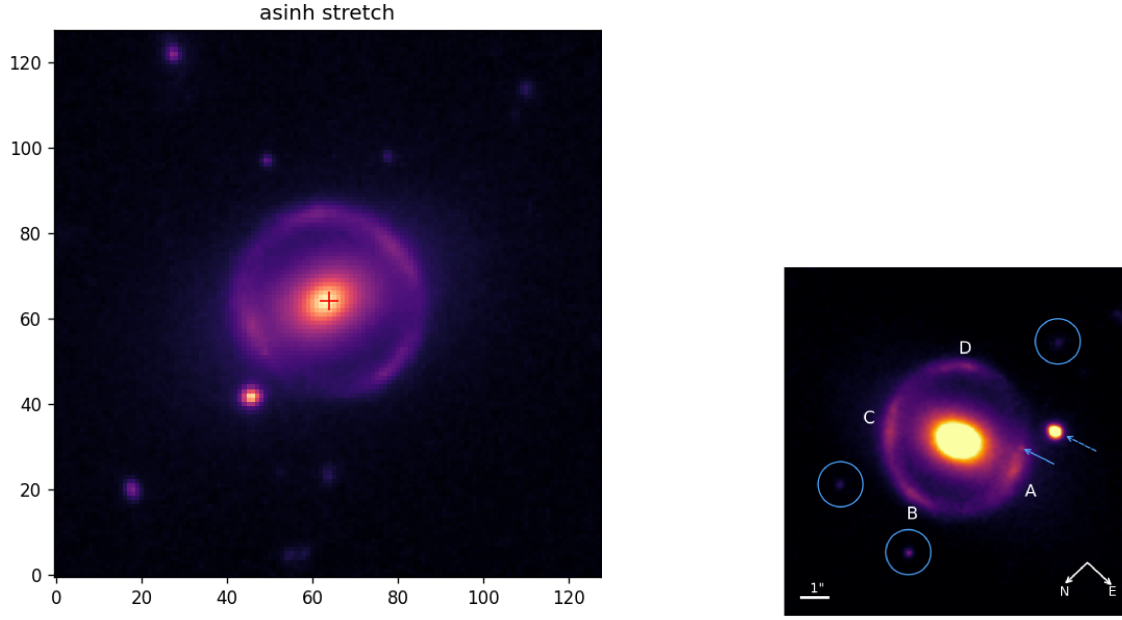


Figure 6: Left: our sky-subtracted F140W cutout (asinh stretch). Right: the published Fig. 7 of Huang et al. (2025) for orientation comparison; note the masked interlopers and companion correspond one-to-one.

Parameter group / parameter	Distribution	Parameters
<i>Mass (EPL + external shear; 8 params)</i>		
θ_E	LogNormal	$\mu = \log 2.5, \sigma = 0.25$
γ_{EPL}	TruncatedNormal	$\mu = 2.0, \sigma = 0.25, [1.0, 2.7]$
e_1^{in}	Normal	$\mu = 0, \sigma = 0.1$
e_2^{in}	Normal	$\mu = 0, \sigma = 0.1$
c_x^{mass}	Normal	$\mu = 0, \sigma = 0.02 \text{ arcsec}$
c_y^{mass}	Normal	$\mu = 0, \sigma = 0.02 \text{ arcsec}$
$\gamma_{\text{ext},1}$	Normal	$\mu = 0, \sigma = 0.05$
$\gamma_{\text{ext},2}$	Normal	$\mu = 0, \sigma = 0.05$
<i>Lens light — main lens (2× Sérsic, 14 params)</i>		
R_{eff} (compact)	LogNormal	$\mu = \log 0.4, \sigma = 0.3$
R_{eff} (extended)	LogNormal	$\mu = \log 2.0, \sigma = 0.3$
n_s	Uniform	$[0.5, 8.0]$
$e_{1,2}^{\text{light}}$	TruncatedNormal	$\mu = 0, \sigma = 0.15, [-0.4, 0.4]$
c_x, c_y	Normal	$\mu = 0, \sigma = 0.02 \text{ arcsec}$
I_e (compact)	LogNormal	$\mu = \log 5.0, \sigma = 0.5$
I_e (extended)	LogNormal	$\mu = \log 2.0, \sigma = 0.5$
<i>Lens light — nearby companion (2× Sérsic, 14 params)</i>		
R_{eff}	LogNormal	$\mu = \log 0.3, 0.6, \sigma = 0.3$
c_x, c_y	Normal	$\mu = (-2.34, -2.86), \sigma = 0.1 \text{ arcsec}$
I_e	LogNormal	$\mu = \log 1.0, 0.5, \sigma = 0.5$
e, n	As main lens	—
<i>Source Sérsic (7 params)</i>		
R_{eff}	LogNormal	$\mu = \log 0.5, \sigma = 0.3$
n_s	Uniform	$[0.5, 6.0]$
$e_{1,2}^{\text{src}}$	TruncatedNormal	$\mu = 0, \sigma = 0.15, [-0.5, 0.5]$
c_x, c_y	Normal	$\mu = 0, \sigma = 0.1 \text{ arcsec}$
I_e	LogNormal	$\mu = \log 2.0, \sigma = 0.5$
<i>Source shapelets ($n_{\text{max}} = 6, 1 + 2 + 28$ params)</i>		
β	LogNormal	$\mu = \log 0.1, \sigma = 0.1$
c_x, c_y	Normal	$\mu = 0, \sigma = 0.05 \text{ arcsec}$
$\text{amp}_i, i = 0, \dots, 27$	Normal	$\mu = 0, \sigma = 5/\sqrt{i+1}$

Lens mass:	$\theta_E \sim \exp(\mathcal{N}(\ln 2.5, 0.25))$
	$\gamma \sim \mathcal{TN}(2, 0.25, 1, 2.7)$
	$e_1, e_2 \sim \mathcal{N}(0, 0.1)$
	$x, y \sim \mathcal{N}(0, 0.05)$
	$\gamma_{\text{ext},1}, \gamma_{\text{ext},2} \sim \mathcal{N}(0, 0.05)$
Lens light:	$R_l \sim \exp(\mathcal{N}(\ln 1, 0.15)) / \exp(\mathcal{N}(\ln 1, 0.15))$
	$n_l \sim \mathcal{U}(0.5, 10) / \mathcal{U}(0.5, 10)$
	$e_{1,1}, e_{1,2} \sim \mathcal{N}(0, 0.15) / \mathcal{N}(0, 0.15)$
	$x_l \sim \mathcal{N}(0, 0.05) / \mathcal{N}(0, 0.10)$
	$y_l \sim \mathcal{N}(0, 0.05) / \mathcal{N}(0, 0.10)$
Source light:	$R_s \sim \exp(\mathcal{N}(\ln 0.25, 0.15))$
	$n_s \sim \mathcal{U}(0.5, 6)$
	$e_{s,1}, e_{s,2} \sim \mathcal{N}(0, 0.15)$
	$x_s, y_s \sim \mathcal{N}(0, 0.1)$
	$\beta_{\text{shp}} \sim \exp(\mathcal{N}(\ln 0.1, 0.1))$
Nearby galaxy:	$x_{\text{shp}}, y_{\text{shp}} \sim \mathcal{N}(0, 0.05)$
	$R_g \sim \exp(\mathcal{N}(\ln 0.4, 0.2)) / \exp(\mathcal{N}(\ln 0.4, 0.2))$
	$n_g \sim \mathcal{U}(0.5, 5) / \mathcal{U}(0.5, 5)$
	$\epsilon_{g,1}, \epsilon_{g,2} \sim \mathcal{TN}(0, 0.15, -0.3, 0.3) / \mathcal{TN}(0, 0.15, -0.3, 0.3)$
	$x_g \sim \mathcal{TN}(3.7, 0.05, 3.55, 3.85) / \mathcal{TN}(3.7, 0.05, 3.55, 3.85)$
	$y_g \sim \mathcal{TN}(0.25, 0.05, 0.1, 0.4) / \mathcal{TN}(0.25, 0.05, 0.1, 0.4)$

Figure 7: Left: our prior table (Appendix A) as built. Right: the published Table 2 priors. The eccentricity priors are fully independent in both — there is no mass–light position–angle tie in the published model.